

2.3 Assignment: Machine Learning vs. Deep Learning

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Machine Learning versus Deep Learning: Choosing the Right Approach for Two Real-World Problems

Machine learning and deep learning both learn from data, but they differ in how features are derived and how much data and computation each one needs. This report examines two examples from the lesson. The first is email spam detection, a simple machine learning problem. The second is medical image analysis, a deep learning problem. For each example I describe a real-world application, explain why the chosen approach fits the problem, and explain why the other approach does not.

Simple Machine Learning: Email Spam Detection

A widely used application is the spam filter that sits on a corporate email gateway and scores every inbound message before it reaches a mailbox. The filter reads a small set of human-understandable signals and decides whether the message is spam or legitimate.

Simple machine learning fits this problem well. The signal lives in a compact set of features that a person can name in advance: the presence of certain words, the reputation of the sender domain, the ratio of links to text, and the length of the message. An algorithm such as logistic regression or a decision tree can learn a reliable boundary from a few thousand labeled messages (Sahami et al., 1998). Training is fast, the model runs cheaply at high volume, and the result is interpretable. A security analyst can see which features flagged a message, which matters when a blocked email has to be justified to the business.

Deep learning is not suitable here. A neural network needs very large labeled datasets and far more computation to match the accuracy that a simple model already reaches on this structured, low-dimensional data (Goodfellow et al., 2016). Its ability to learn features automatically adds little value, because the useful features are already known and easy to extract. The network would also hide its reasoning inside layers of weights, and that loss of transparency is a real cost

in a security setting where decisions must be explained.

Deep Learning: Medical Image Analysis

A well-established application is the automated screening of retinal photographs to detect diabetic retinopathy, a leading cause of blindness. A trained system grades the severity of disease in each image so that patients who need a specialist are prioritized quickly (Gulshan et al., 2016).

Deep learning fits this problem well. The input is raw pixels that contain complex spatial patterns, such as tiny hemorrhages, lesions, and subtle changes in texture spread across the whole image. No person can write down a short list of features that captures all of this reliably. A convolutional neural network learns a hierarchy of features directly from labeled images, moving from simple edges to lesion shapes, and it can reach expert-level accuracy when trained on enough data (Gulshan et al., 2016).

Simple machine learning is not suitable here. A traditional algorithm cannot consume millions of raw pixels directly; it depends on features that an engineer defines ahead of time. Hand-crafting features that describe every relevant lesion is impractical, and it would miss patterns that humans do not think to look for. Accuracy would fall well short of what a clinical screening tool requires.

Conclusion

The right choice is driven by the structure of the data and the supply of labeled examples. When the important features are few, already known, and worth keeping visible, simple machine learning is faster, cheaper, and easier to trust. When the data is high-dimensional and the patterns are too complex to describe by hand, deep learning earns its added cost. Matching the method to the problem, rather than defaulting to the most powerful tool available, is what makes a solution effective.

References

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